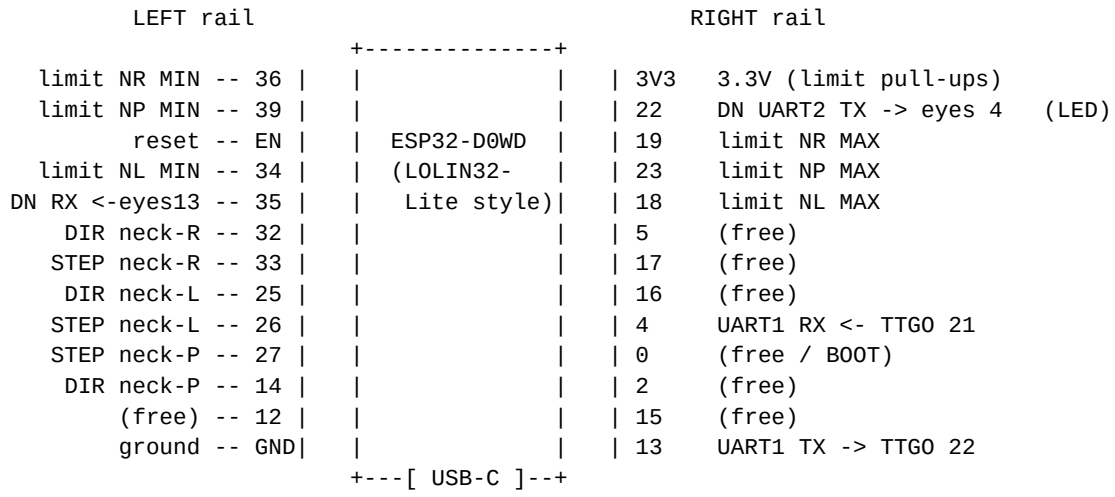
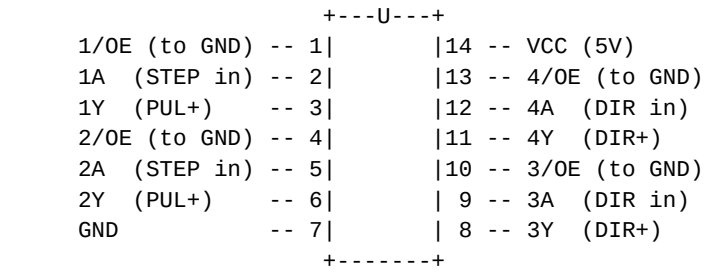
j4_stepper_neck -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM



SN74AHCT125N x3, one per driver (DIP-14, top view, notch at the top)



Each chip: pins 2+5 tied <- STEP, pins 3+6 tied -> PUL+ ;
pins 9+12 tied <- DIR, pins 8+11 tied -> DIR+
U1 neck-left: STEP 26, DIR 25 U2 neck-right: STEP 33, DIR 32
U3 neck-pivot: STEP 27, DIR 14
GPIO 21 is NOT present on this board, which is why the downstream TX is on 22.
Neck STEP/DIR pins pass through the 74AHCT125 buffers (3.3V -> 5V) into the DM556TE PUL+/DIR+ inputs (PUL-/DIR- to GND). 100nF across each chip's VCC/GND; all /OE pins to GND.
Limits: one NC switch per endpoint. MIN ends (34/36/39) need external 10k pull-ups to 3.3V; MAX ends (18/19/23) use the internal pull-ups.

GPIO	Function
26 / 25	neck-left STEP / DIR -> 74AHCT125 U1 -> DM556TE PUL+ / DIR+
33 / 32	neck-right STEP / DIR -> 74AHCT125 U2 -> DM556TE PUL+ / DIR+
27 / 14	neck-pivot STEP / DIR -> 74AHCT125 U3 -> DM556TE PUL+ / DIR+
4 / 13	Upstream UART1 RX / TX -- to j4_receiver: RX from TTGO GPIO 21, TX to TTGO GPIO 22
22 / 35	Downstream UART2 TX / RX -- to j4_stepper_eyes: TX to eyes GPIO 4, RX from eyes GPIO 13
34	neck-left MIN limit (input-only, external 10k pull-up)
36	neck-right MIN limit (input-only, external 10k pull-up)
39	neck-pivot MIN limit (input-only, external 10k pull-up)
18	neck-left MAX limit (internal pull-up)
19	neck-right MAX limit (internal pull-up)
23	neck-pivot MAX limit (internal pull-up)